

Industry: Healthcare / Medical Signal Processing

Title: ECG Signal Classification for Arrhythmia Detection Using LSTM and GRU Networks

Difficulty Level: Advanced

1. Industry-Oriented Problem Description

Cardiovascular diseases remain one of the leading causes of death worldwide.

Electrocardiogram (ECG) monitoring is widely used to detect irregular heart rhythms (arrhythmias). Traditionally, cardiologists manually analyze ECG waveforms to detect abnormalities such as atrial fibrillation, ventricular tachycardia, and premature contractions. However, manual interpretation is time-consuming and subject to inter-observer variability.

The objective of this project is to design a deep learning-based time-series classification system

that analyzes ECG signal sequences and classifies them into normal or specific arrhythmia categories. ECG signals are continuous temporal waveforms, making this a sequential learning problem rather than a static feature classification task.

Students must preprocess ECG signals into fixed-length sequences and experiment with recurrent neural network architectures such as LSTM and GRU. They may optionally compare performance with a 1D CNN model. The final system should output arrhythmia class probabilities and provide evaluation using medically relevant metrics.

2. Technical Direction Clues

Load ECG signal dataset (e.g., MIT-BIH or synthetic data).

Segment signals into fixed-length time windows.

Normalize signal amplitude values.

Apply noise filtering techniques if required.

Construct input tensor: (samples, time_steps, features).

Train LSTM and GRU models for classification.

Optionally compare with 1D CNN model.

Use Categorical Cross-Entropy loss for multi-class classification.

Evaluate using Accuracy, F1-score, Sensitivity, and Specificity.

3. Sample Database Schema

Table: Patients

patient_id | INT | Unique identifier

age | INT | Age of patient
gender | VARCHAR | Male / Female
diagnosis_label | VARCHAR | Normal / AFib / VTach / PVC

Table: ECG_Records

record_id | INT | Unique identifier
patient_id | INT | Foreign key
signal_sequence | FLOAT ARRAY | ECG waveform values
sampling_rate | INT | Samples per second
record_duration | FLOAT | Duration in seconds

4. Required Metadata / Artifacts

Signal preprocessing pipeline description

Sequence window size selection

Noise filtering techniques applied

Model architecture diagrams

Hyperparameters (learning rate, batch size, epochs)

Loss function used

Confusion matrix

Classification report

Model comparison table (LSTM vs GRU vs 1D CNN)

ROC curves per class

5. Model Suitability Explanation

LSTM: Captures long-term dependencies in ECG waveforms.

GRU: Efficient sequential model suitable for real-time monitoring.

1D CNN: Captures local waveform patterns.

MLP alone is insufficient due to loss of temporal structure.

6. Expected Outcomes

Trained arrhythmia classification model

Performance comparison between LSTM, GRU, and 1D CNN

Accurate arrhythmia detection metrics

Medical interpretation of model sensitivity and specificity

Deployment-ready ECG monitoring inference system